Step 1: Provisioning the Amazon S3 Bucket

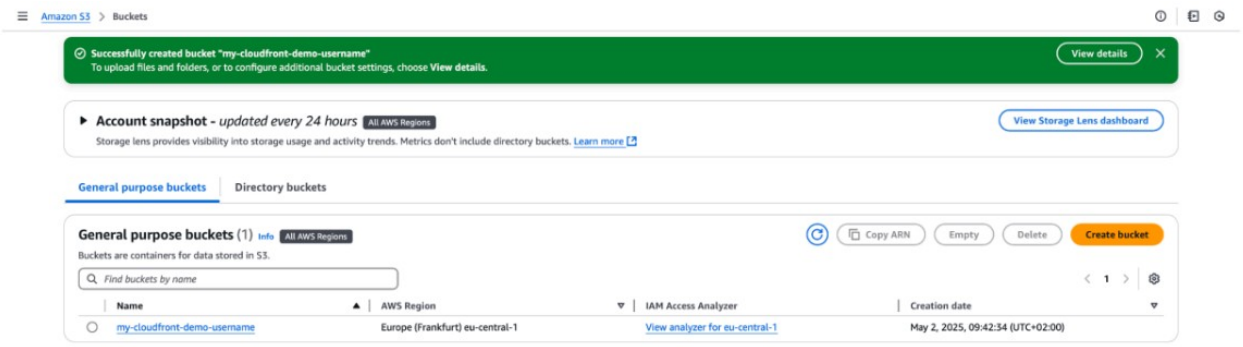
The first phase of this infrastructure deployment involves creating an **Amazon S3 (Simple Storage Service)** bucket. In this lab, the bucket serves as the **Origin** for a CloudFront distribution, providing a scalable and durable storage solution for web content.

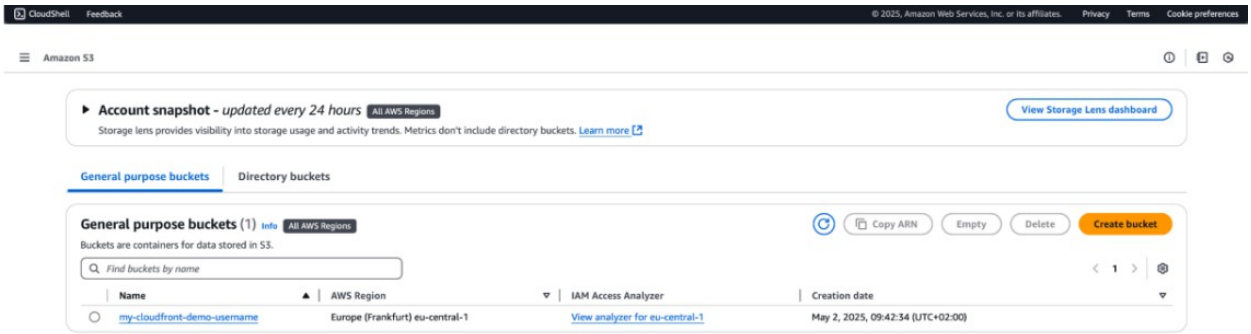
Technical Implementation:

- **Bucket Name:** `my-cloudfront-demo-username` (Must be globally unique and DNS-compliant).
- **Region:** `eu-central-1` (Frankfurt). Choosing a region close to the target audience or compliant with data residency regulations is a key architectural decision.
- **Bucket Type:** General Purpose.

Cybersecurity Best Practices (Professional Insight):

1. **Block Public Access (BPA):** By default, S3 buckets should have "Block all public access" enabled. This prevents accidental data exposure—one of the most common causes of cloud data breaches.
2. **Principle of Least Privilege (PoLP):** Access to this bucket is strictly managed via IAM (Identity and Access Management) roles and Bucket Policies, ensuring only authorized services (like CloudFront via OAC/OAI) can retrieve the data.
3. **Encryption:** For a production-ready environment, I would enable **SSE-S3** or **SSE-KMS** to ensure data is encrypted at rest.





Step 2: Configuring S3 Public Access and Bucket Policy

To allow **Amazon CloudFront** to retrieve and serve content from the S3 origin, the bucket permissions must be configured to permit read access. This step involves adjusting the "Block Public Access" settings and defining a granular JSON bucket policy.

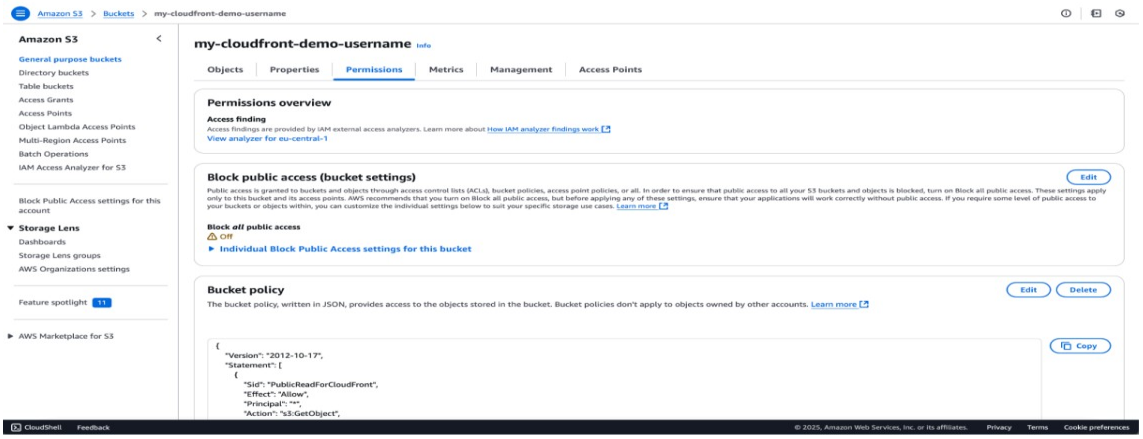
Technical Implementation:

1. **Adjusting Block Public Access (BPA):** Navigated to the **Permissions** tab of the bucket.
    - Disabled the **"Block all public access"** setting. This is a prerequisite for applying a bucket policy that allows external read access.
2. **Defining the Bucket Policy:** Applied a JSON-based resource policy to grant `s3:GetObject` permissions.
    - **Effect:** `Allow`
    - **Principal:** (Configured for public read access in this lab environment).
    - **Resource:** `arn:aws:s3:::my-cloudfront-demo-username/*`

Cybersecurity Perspective: Risk Assessment & Mitigation

In a real-world production environment, disabling "Block Public Access" is generally considered a high-risk configuration. Here is how I approach this from a security standpoint:

- **The Risk:** Using a wildcard principal (`*`) in a bucket policy can lead to unauthorized data exposure if not managed correctly.
- **The Lab Context:** This configuration was used to demonstrate the basic integration between S3 and CloudFront.
- **Professional Best Practice (The "Pro" Move):** For a production-grade deployment, I would implement **Origin Access Control (OAC)**. This allows the bucket to remain **private** (with "Block Public Access" enabled) while granting access *only* to the specific CloudFront distribution service principal. This adheres to the **Principle of Least Privilege (PoLP)** and prevents users from bypassing the CDN to access the origin directly.



Step 3: Object Ingestion and Asset Management

With the bucket infrastructure and permissions in place, the next phase is the ingestion of static assets. This step demonstrates the ability to manage objects within the S3 environment that will eventually be served through a Content Delivery Network (CDN).

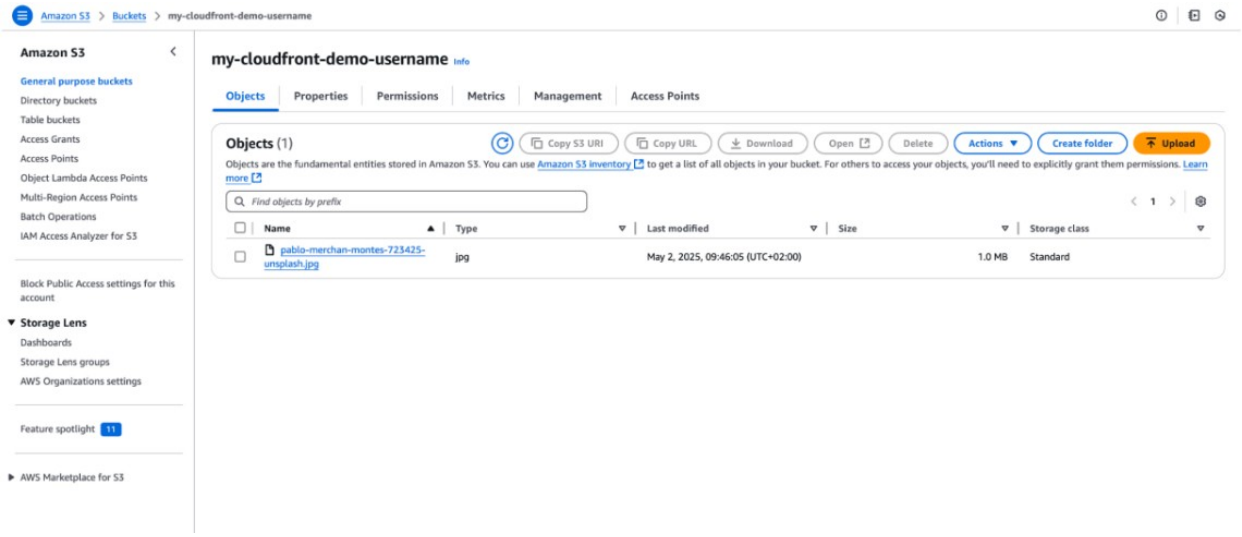
Technical Implementation:

- **Navigation:** Accessed the **Objects** tab within the `my-cloudfront-demo-username` bucket.
- **Action:** Successfully uploaded a sample asset (Image file: `pablo-merchan-montes...jpg`).
- **Storage Class:** Assigned to **S3 Standard**. This class provides 99.999999999% (11 nines) of durability, making it ideal for high-availability web content.

Cybersecurity & Governance (The Professional Edge):

From a security perspective, uploading a file isn't just a "drag-and-drop" task. In a professional environment, I consider the following:

1. **Data Integrity (Checksums):** To ensure that the file was not corrupted or tampered with during the upload process, I utilize the **ETag (MD5 hash)** or provide a **SHA-256 checksum**. This guarantees the integrity of the source data.
2. **Metadata Scrubbing:** Before uploading public-facing assets, it is a best practice to strip sensitive **EXIF metadata** (such as GPS coordinates or camera serial numbers) to prevent information leakage to potential attackers.
3. **Versioning & Immutability:** In a production setup, I would enable **S3 Versioning** or **Object Lock**. This provides a defense-in-depth layer against ransomware or accidental deletion by keeping a history of objects or making them "WORM" (Write Once, Read Many).



Step 4: Provisioning an Amazon CloudFront Distribution

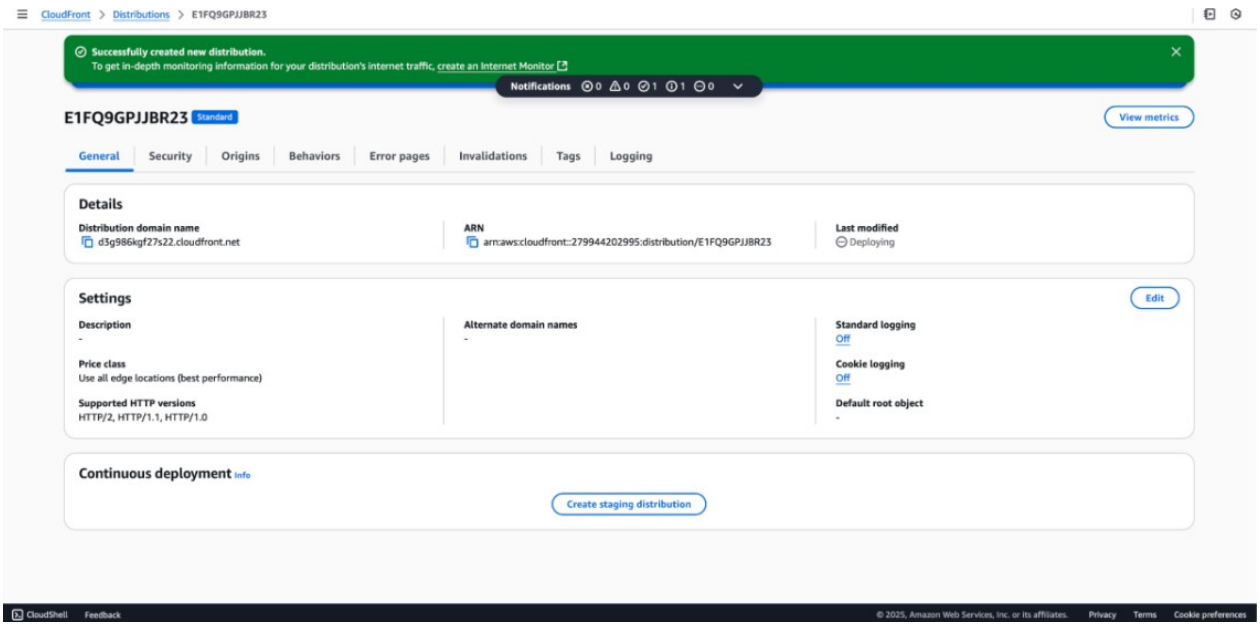
To deliver the content stored in the S3 bucket securely and with low latency, I provisioned an **Amazon CloudFront Distribution**. This creates a global **Content Delivery Network (CDN)** that caches content at **Edge Locations**, reducing the load on the origin server and improving the end-user experience.

Technical Implementation:

- **Distribution Domain Name:** `d3g986kgf27s22.cloudfront.net` (The secure endpoint for users).
- **Price Class:** `Use all edge locations (best performance)`. This ensures the lowest possible latency by utilizing AWS's global network of Points of Presence (PoPs).
- **Supported Protocols:** Enabled `HTTP/2, HTTP/1.1, and HTTP/1.0` to ensure broad compatibility while optimizing performance for modern, secure browsers.
- **Status:** Initial deployment and global propagation across the AWS network.

Cybersecurity Highlights (Infrastructure Hardening):

1. **DDoS Mitigation (AWS Shield Standard):** By using CloudFront, this infrastructure is automatically protected by **AWS Shield Standard**. This provides always-on detection and automatic inline mitigations against common Layer 3 and 4 infrastructure attacks.
2. **Encryption in Transit:** CloudFront is configured to enforce **HTTPS**, ensuring that data transferred between the end-user and the Edge Location is encrypted using modern **TLS** protocols.
3. **Reducing the Attack Surface:** By directing all traffic through CloudFront, I can effectively hide the direct S3 bucket URL from the public internet. This ensures that the origin is not directly exposed to potential attackers.
4. **WAF Integration Readiness:** This distribution is designed to integrate seamlessly with **AWS WAF (Web Application Firewall)**. In a production scenario, I would implement custom WAF rules to filter out malicious traffic, such as SQLi or XSS, at the edge before it reaches the source.



Step 5: Configuring Cache Behaviors and Access Control

The **Behavior** settings in CloudFront act as the traffic controller for the distribution. By fine-tuning these settings, I ensured that the delivery of content is both highly performant and strictly controlled according to the **Principle of Least Privilege**.

Technical Implementation:

- **Path Pattern:** `Default (*)` — All requests to the distribution follow these rules unless specific path overrides are created.
- **Allowed HTTP Methods:** Restricted to `GET, HEAD`. This is a deliberate security choice to prevent unauthorized users from attempting state-changing operations (like `POST`, `PUT`, or `DELETE`) at the edge.
- **Cache Policy:** Implemented the `Managed-CachingOptimized` policy. This optimizes the **Time-To-Live (TTL)** settings, ensuring that content is cached effectively at Edge Locations to minimize latency and reduce origin (S3) costs.
- **Viewer Protocol Policy:** Currently set to `HTTP and HTTPS` for initial testing, with the recommendation to enforce `Redirect HTTP to HTTPS` for production hardening.

Cybersecurity Analysis: Method Filtering & Origin Protection

In a security-focused environment, we don't just "enable" a CDN; we "harden" it:

1. **Method Whitelisting:** By restricting allowed methods to only `GET` and `HEAD`, I have effectively eliminated entire classes of web-based attacks that rely on other HTTP verbs.
2. **Origin Shielding:** CloudFront acts as a buffer. Even if a malicious actor attempts to flood the site with requests, the **CachingOptimized** policy ensures that most traffic is served from the AWS global cache, preventing the S3 origin from being overwhelmed or incurring massive costs.

Step 5.1: Final Verification (System Integration Test)

To verify the end-to-end integration, I accessed the static asset using the unique CloudFront Domain Name.

Results:

- **Success URL:** `https://d3g986kgf27s22.cloudfront.net/pablo-merchan-montes-723425-unsplash.jpg`
- **Outcome:** The asset was successfully retrieved and rendered via the global CDN, confirming that the S3 Bucket Policy, CloudFront Origin settings, and Cache Behaviors are correctly synchronized.

CloudFront > Distributions > E1FQ9GPJJBR23

E1FQ9GPJJBR23 Standard

View metrics

General | Security | Origins | Behaviors | Error pages | Invalidations | Tags | Logging

Behaviors

Save | Move up | Move down | Edit | Delete | Create behavior

Filter behaviors by property or value

Preced...	Path pattern	Origin or origin gr...	Viewer protocol po...	Cache policy name	Origin request poli...	Trusted key groups	Trusted signers	Response ...
0	Default (*)	my-cloudfront-dem...	HTTP and HTTPS	Managed-CachingOptimized	-	-	-	-

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CloudFront > Distributions > E1FQ9GPJJBR23 > Edit behavior

Edit behavior

Settings

Path pattern Info

Default (*)

Origin and origin groups

my-cloudfront-demo-username.s3.eu-central-1.amazonaws.com

Compress objects automatically Info

- No
- Yes

Viewer

Viewer protocol policy

- HTTP and HTTPS
- Redirect HTTP to HTTPS
- HTTPS only

Allowed HTTP methods

- GET, HEAD
- GET, HEAD, OPTIONS
- GET, HEAD, OPTIONS, PUT, POST, PATCH, DELETE

Restrict viewer access

If you restrict viewer access, viewers must use CloudFront signed URLs or signed cookies to access your content.

- No
- Yes

Cache key and origin requests

We recommend using a cache policy and origin request policy to control the cache key and origin requests.

- Cache policy and origin request policy (recommended)
- Legacy cache settings

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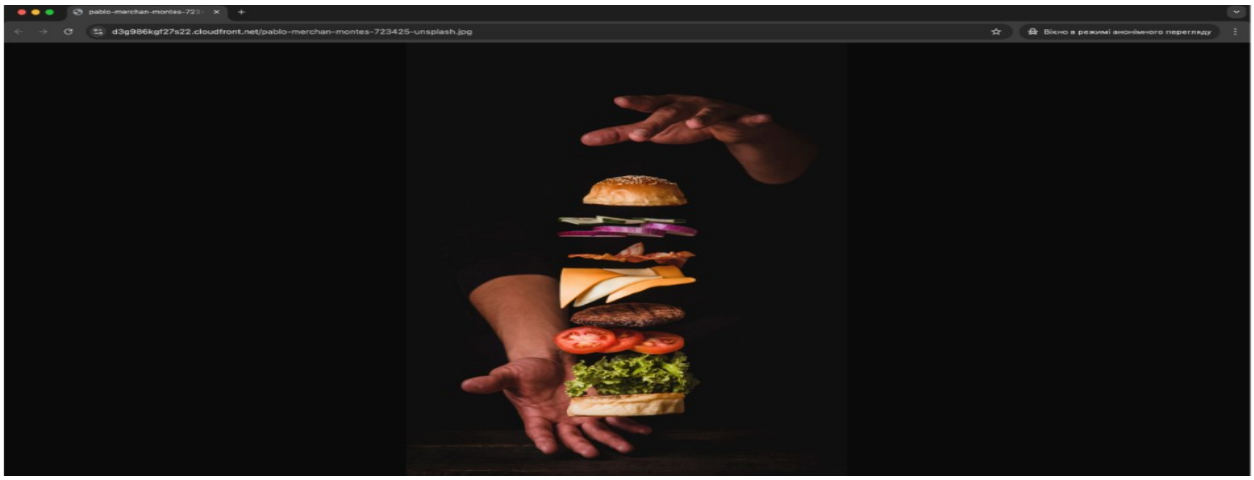
d3g986kgf27o22.cloudfront.net/pablo-merchan-montes-723425-unsplash.jpg



Step 6: Resource Decommissioning and Environment Cleanup

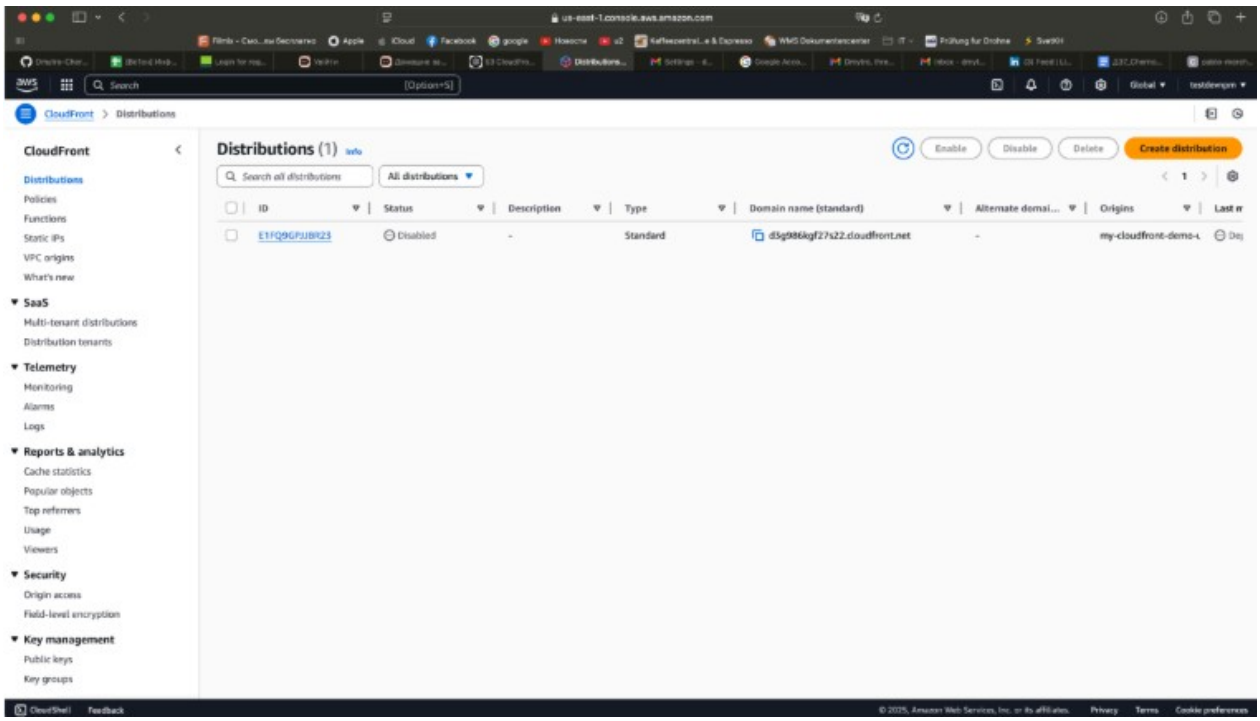
The final phase of the cloud lifecycle is the secure decommissioning of resources. Proper cleanup is a critical skill in cloud management, ensuring that no "ghost" resources remain active, which could lead to unnecessary costs or security vulnerabilities.

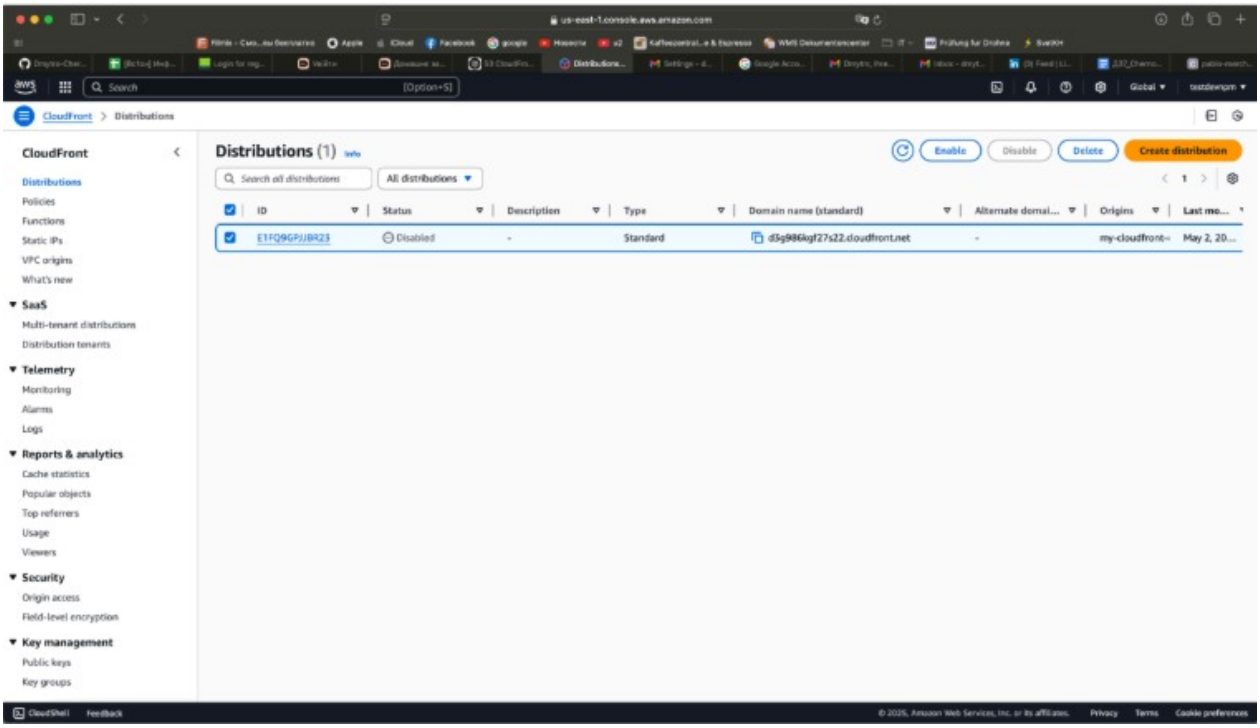
Technical Implementation:

1. **CloudFront Distribution Cleanup:**
    - **Action:** Transitioned the distribution to a **Disabled** state. AWS requires a distribution to be disabled before it can be deleted to prevent accidental service disruption.
    - **Action:** Once disabled, the distribution was permanently **deleted** from the AWS global network.
2. **Amazon S3 Bucket Cleanup:**
    - **Action:** Emptied the `my-cloudfront-demo-username` bucket. S3 prevents the deletion of non-empty buckets as a data loss prevention measure.
    - **Action:** Successfully deleted the bucket infrastructure.

Cybersecurity & Operational Excellence (The "Pro" Mindset):

- **Cost Optimization (FinOps):** Effective cloud security includes financial security. By promptly removing resources after a lab or project, I demonstrate an understanding of the **Pay-as-you-go** model and responsible budget management.
- **Reducing the Attack Surface:** Every active endpoint is a potential entry point for an attacker. Removing unused distributions and buckets is a key part of **Security Hygiene** — if a resource doesn't exist, it cannot be compromised.
- **Stale Resource Management:** Deleting resources prevents "Shadow IT" or stale environments that might be forgotten and left unpatched or unmonitored over time.




us-east-1.console.aws.amazon.com
CloudFront > Distributions
CloudFront
Distributions
Policies
Functions
Static IPs
VPC origins
What's new
SaaS
Multi-tenant distributions
Distribution tenants
Telemetry
Monitoring
Alarms
Logs
Reports & analytics
Cache statistics
Popular objects
Top referrers
Usage
Viewers
Security
Origin access
Field-level encryption
Key management
Public keys
Key groups
Distributions (1) Info
Enable
Disable
Delete
Create distribution
Search all distributions
All distributions
ID
Status
Description
Type
Domain name (standard)
Alternate domai...
Origins
Last mo...
E1FQ9GPJJBR23
Disabled
Standard
d3g986kgf27s22.cloudfront.net
my-cloudfront-
May 2, 20...
CloudShell
Feedback
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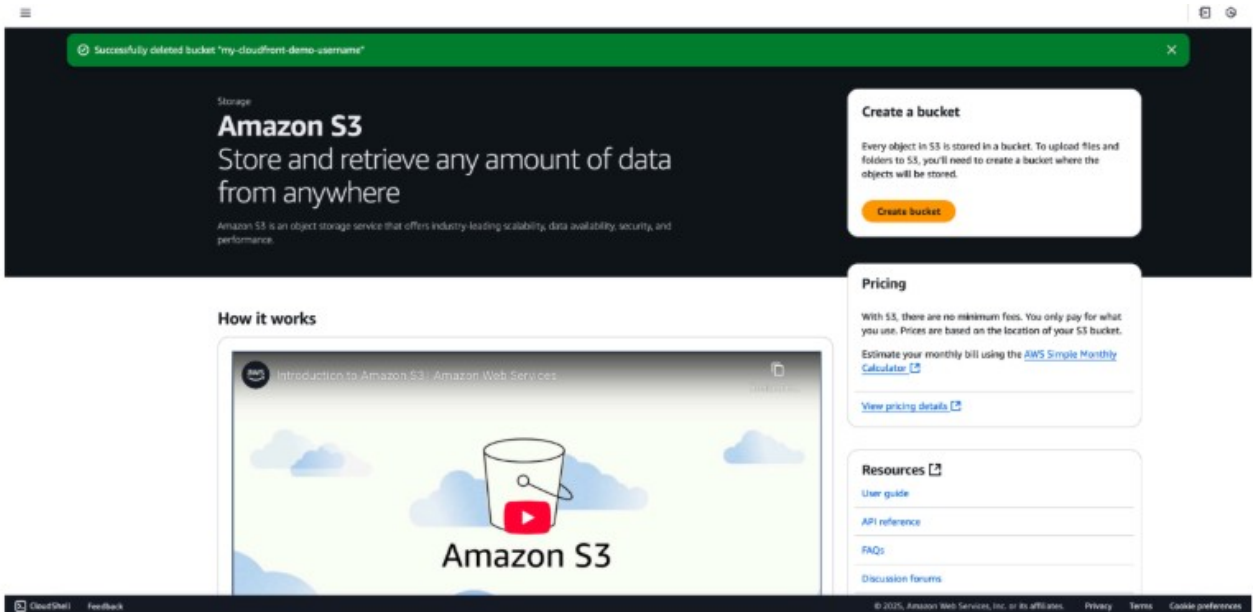
Successfully deleted bucket "my-cloudfront-demo-username"
Storage
Amazon S3
Store and retrieve any amount of data from anywhere
Amazon S3 is an object storage service that offers industry-leading scalability, data availability, security, and performance.
Create a bucket
Every object in S3 is stored in a bucket. To upload files and folders to S3, you'll need to create a bucket where the objects will be stored.
Create bucket
How it works
Introduction to Amazon S3 | Amazon Web Services
Amazon S3
Pricing
With S3, there are no minimum fees. You only pay for what you use. Prices are based on the location of your S3 bucket.
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